

Mind the Margin: Safety- and Physical-Aware Planning for End-to-End Autonomous Driving

Qi Xiong¹ , Jinlai Zhang², Chuan Hu³, Hao Chen², and Lin Hu²

Abstract—While end-to-end autonomous driving has demonstrated remarkable progress, existing frameworks frequently struggle with physical infeasibility, inadequate safety margins, and severe terminal drift over long planning horizons. These limitations stem from a fundamental disconnect between purely data-driven imitation and the stringent kinematic and safety constraints required in real-world driving. To bridge this gap, we propose MarginAD, a novel safety- and physical-aware planning framework that seamlessly integrates explicit physical priors and safety constraints into the end-to-end optimization pipeline. Specifically, MarginAD introduces physics-aware kinematic guidance that injects trajectory-level kinematic cues directly into the planning queries, promoting dynamically feasible motion generation. We then apply a safety-margin-aware collision loss to penalize unsafe proximity to predicted surrounding agents, and further use horizon-aware regression weighting to suppress terminal drift. Compared with the strong MomAD baseline, MarginAD reduces the average L2 error on nuScenes from 0.60 m to 0.55 m, the average collision rate from 0.09% to 0.08%, and the average TPC from 0.54 m to 0.47 m; on the more challenging Shift-nuScenes subset, it further reduces the average L2 error from 1.54 m to 0.75 m and the average collision rate from 0.38% to 0.19%. The code is available at <https://github.com/xiongqi123123/MarginAD>.

Index Terms—end-to-end autonomous driving, motion planning, safety-aware planning, physics-constrained planning.

I. INTRODUCTION

Autonomous driving has progressively shifted from modular pipelines toward end-to-end frameworks, which now form a central research direction in the field [1]. From early behavior-cloning policies [2] to integrated sensor-fusion systems [3] and trajectory-based neural planners [4], this line is appealing because it optimizes driving behavior directly from sensory inputs. However, current trajectory-based planners are still largely supervised by position-level imitation, so they may remain close to the expert path on average while still producing behavior that is weak in physical feasibility, safety-margin preservation, and long-horizon stability. This mismatch becomes especially problematic in interactive scenarios, where

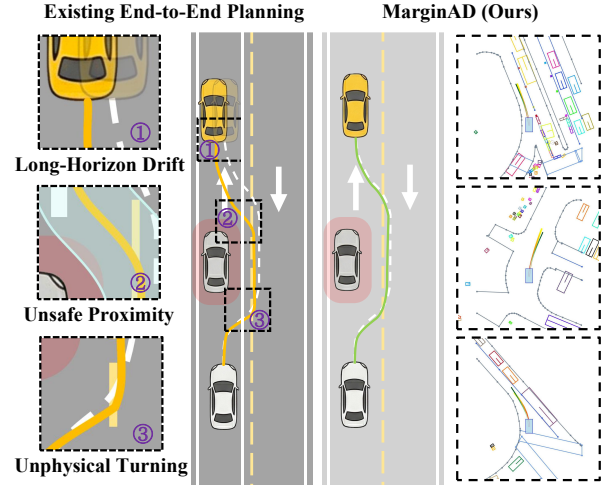


Fig. 1. Existing end-to-end planners often suffer from long-horizon drift, unsafe proximity, and unphysical turning. In contrast, MarginAD effectively mitigates these issues by generating temporally stable, safety-aware, and kinematically feasible trajectories.

small geometric errors can quickly lead to unsafe or physically implausible maneuvers.

Recent planning-oriented frameworks have substantially advanced trajectory prediction quality. UniAD [5], VAD [6], SparseDrive [7], and MomAD [8] improve unified planning, scene representation, efficiency, and temporal consistency, yet they still formulate ego planning as data-driven trajectory regression with little explicit regularization for vehicle kinematics or safety margins. Reliable end-to-end planning therefore requires not only stronger scene representation, but also direct physical and safety constraints within the planner itself.

To address these limitations, we propose MarginAD, a framework that injects vehicle kinematic constraints and safety margins into trajectory generation while preserving the architectural efficiency of the backbone. MarginAD refines the planning head through physics-aware kinematic guidance, a safety-margin-aware collision loss, and horizon-aware regression weighting to jointly improve kinematic feasibility, safety-margin preservation, and long-horizon stability.

II. RELATED WORK

End-to-End Autonomous Driving. Predicting future ego trajectories has become a standard planning interface in end-to-end autonomous driving [4]–[13]. Early systems such as the interpretable neural motion planner [4] and MP3 [9] showed that planning can be learned jointly with scene understanding, while UniAD [5] and VAD [6] established unified query-based

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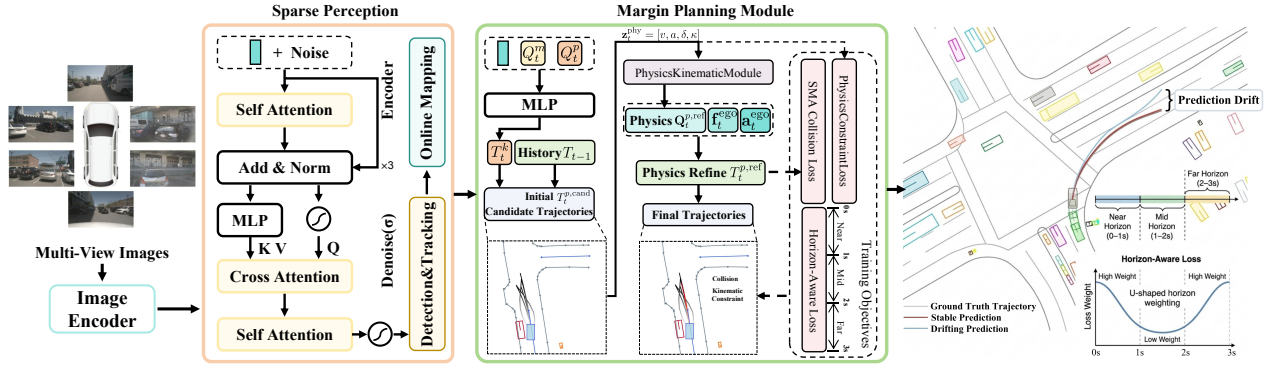


Fig. 2. **The overall architecture of MarginAD.** Scene features from multi-view images drive perception, motion forecasting, and ego planning; the planning query is then refined with physics-aware kinematic cues and optimized under physical supervision, a safety-margin-aware collision loss, and horizon-aware regression weighting.

and vectorized driving stacks. More recent methods, including SparseDrive [7], MomAD [8], and PLUTO [10], improve efficiency, temporal consistency, and imitation quality, but they still mainly strengthen data-driven trajectory regression itself. MarginAD addresses a complementary gap by injecting physical feasibility and safety-margin preservation directly into planning refinement.

Physical Feasibility in End-to-End Planning. Physical feasibility is a long-standing requirement in motion planning and control, as reflected in constrained trajectory optimization, MPC-style planning, and related controller designs [14]–[16]. In end-to-end driving, recent work has started to introduce physically grounded structure through physics-informed objectives or occupancy-based planning formulations [17], [18]. However, in trajectory-based planners, feasibility cues are still often implicit in regression losses or deferred to downstream correction. GuideFlow [12] addresses this by enforcing explicit safety and physical constraints inside a flow-matching generation process, removing the need for a separate refinement stage. MarginAD instead moves physical reasoning into the planning head itself through kinematic query refinement and direct physical supervision.

Safety Modeling in End-to-End Planning. Safety has been introduced into end-to-end planning through semantic cost maps, multi-future evaluation, safety-constrained optimization, and interaction-aware prediction modules [19]–[22]. Recent work such as SafeDrive [23] makes this reasoning more explicit with trajectory-conditioned safety analysis over future timesteps. MarginAD follows a more direct planner-level route: instead of adding a separate safety reasoning stack, we inject a differentiable safety-margin penalty into trajectory refinement and couple it with physically grounded query enhancement. This design lets the safety signal directly regularize the final ego trajectory while preserving the efficiency of a standard trajectory-based planner.

III. METHOD

Figure 2 shows the overall pipeline. We follow the standard trajectory-based end-to-end setting, where multi-view images are encoded into shared scene features for detection, tracking, mapping, motion forecasting, and ego planning. MarginAD keeps this pipeline and backbone unchanged and focuses on planning refinement, where the final ego trajectory most di-

rectly determines physical feasibility, safety margin, and long-horizon stability. Starting from the first-stage planning output, it refines the planning query with reconstructed kinematic cues under physical supervision (§III-A), adds a confidence-aware safety-margin penalty against unsafe proximity (§III-B), and reweights the regression loss along the horizon to suppress terminal drift (§III-C).

A. Physics-Aware Kinematic Guidance

Position-level imitation alone can still produce unstable motion, such as abrupt turning or implausible curvature changes. To address this issue, MarginAD reconstructs trajectory-level kinematic cues from the first-stage prediction, feeds them back into the planning query, and applies direct physical supervision to the refined trajectory.

Given the predicted displacement sequence $\{\Delta\tau_t\}_{t=1}^T$, we first reconstruct the future position sequence

$$\tau_t = \sum_{i=1}^t \Delta\tau_i, \quad \tau_t = (x_t, y_t), \quad t = 1, \dots, T. \quad (1)$$

We then compute velocity by finite differences:

$$\mathbf{v}_t = \frac{\tau_{t+1} - \tau_t}{\Delta t}, \quad t = 1, \dots, T-1, \quad (2)$$

where $\Delta t = 0.5$ s. The heading angle is

$$\psi_t = \text{atan2}(v_t^y, v_t^x), \quad t = 1, \dots, T-1. \quad (3)$$

Using the discrete bicycle-model relation, we estimate the steering-related curvature and steering angle by

$$\kappa_t^{\text{steer}} = \frac{\text{wrap}(\psi_{t+1} - \psi_t)}{\Delta t \cdot \max(\|\mathbf{v}_t\|_2, \epsilon)}, \quad (4)$$

$$\delta_t = \arctan(L\kappa_t^{\text{steer}}), \quad t = 1, \dots, T-2,$$

where L is the wheelbase, ϵ is a small constant for numerical stability, and $\text{wrap}(\cdot)$ maps angle differences to $(-\pi, \pi]$. We further measure the geometric bending of the trajectory through path curvature. We first define

$$\Delta x_t = x_{t+1} - x_t, \quad (5)$$

$$\Delta y_t = y_{t+1} - y_t, \quad t = 1, \dots, T-1,$$

$$\Delta^2 x_t = \Delta x_{t+1} - \Delta x_t,$$

$$\Delta^2 y_t = \Delta y_{t+1} - \Delta y_t, \quad (6)$$

$$t = 1, \dots, T-2.$$

and then compute

$$\kappa_t^{\text{path}} = \frac{|\Delta x_t \Delta^2 y_t - \Delta y_t \Delta^2 x_t|}{\max\{(\Delta x_t^2 + \Delta y_t^2)^{3/2}, \epsilon\}}, \quad t = 1, \dots, T-2. \quad (7)$$

The two curvature terms serve different roles: κ_t^{steer} captures the instantaneous steering demand implied by the heading change, while κ_t^{path} measures the geometric bending of the predicted path itself. We summarize these cues as

$$\mathbf{s}_{\text{phys}} = \left[\|\mathbf{v}_1\|_2, \|\mathbf{v}_{T-1}\|_2, \bar{v}, \bar{\delta}, \bar{\Delta\delta}, \bar{\kappa}^{\text{path}}, \bar{\Delta\kappa}^{\text{path}} \right], \quad (8)$$

where

$$\bar{v} = \frac{1}{T-1} \sum_{t=1}^{T-1} \|\mathbf{v}_t\|_2, \quad \bar{\delta} = \frac{1}{T-2} \sum_{t=1}^{T-2} |\delta_t|, \quad (9)$$

$$\bar{\Delta\delta} = \frac{1}{T-3} \sum_{t=1}^{T-3} |\delta_{t+1} - \delta_t|, \quad \bar{\kappa}^{\text{path}} = \frac{1}{T-2} \sum_{t=1}^{T-2} \kappa_t^{\text{path}}, \quad (10)$$

$$\bar{\Delta\kappa}^{\text{path}} = \frac{1}{T-3} \sum_{t=1}^{T-3} |\kappa_{t+1}^{\text{path}} - \kappa_t^{\text{path}}|. \quad (11)$$

The PhysicsKinematicModule encodes $\mathbf{s}_{\text{phys}}$ and fuses it back into the planning query.

The physics-enhanced query is obtained through a gated residual design:

$$Q_{\text{fused}} = \sigma(\alpha) Q_{\text{orig}} + (1 - \sigma(\alpha)) \text{MLP}_{\text{fusion}}([Q_{\text{orig}}, E_{\text{phys}}]), \quad (12)$$

where E_{phys} denotes the encoded kinematic features and α is a learnable scalar. This design lets the planner adaptively balance the original perception-driven query against the physics-informed refinement, instead of forcing the physical cues to overwrite the scene evidence in every case.

We also apply explicit constraint-violation penalties to the refined trajectory. To focus the physical supervision on lateral stability and path smoothness, the final configuration penalizes steering-angle and curvature violations:

$$\mathcal{L}_{\text{phys}} = \lambda_{\delta} \mathcal{L}_{\text{steer}} + \lambda_{\kappa} \mathcal{L}_{\text{curv}}, \quad (13)$$

with

$$\mathcal{L}_{\text{steer}} = \frac{1}{T-2} \sum_{t=1}^{T-2} \max(0, |\delta_t| - \delta_{\text{max}}), \quad (14)$$

$$\mathcal{L}_{\text{curv}} = \frac{1}{T-2} \sum_{t=1}^{T-2} \max(0, \kappa_t^{\text{path}} - \kappa_{\text{max}}), \quad (15)$$

where δ_{max} and κ_{max} are predefined vehicle limits. In this way, the planner is guided both by physics-aware query refinement and by direct supervision on kinematic feasibility.

B. Safety-Margin-Aware Collision Loss

Imitation loss encourages the ego trajectory to follow the expert path, but it does not directly penalize trajectories that pass too close to surrounding agents. This limitation becomes more visible in interactive scenes such as obstacle bypassing and lane changes, where a trajectory can remain close to the ground truth while still leaving insufficient safety margin. MarginAD addresses this issue with a differentiable safety-margin-aware collision loss on the refined ego trajectory.

For each training sample, we first select the positive refined ego mode and reconstruct its absolute trajectory $\tau_{\text{ego}} \in \mathbb{R}^{T \times 2}$ by cumulative summation. For surrounding agents, we take the highest-confidence mode predicted by the motion head for each detected instance and convert the predicted displacements

into absolute trajectories $\tau_{\text{agent}} \in \mathbb{R}^{N \times T \times 2}$ using the detection anchors. The center distance between the ego trajectory and each agent trajectory is

$$d_{n,t} = \|\tau_{\text{ego},t} - \tau_{\text{agent},n,t}\|_2. \quad (16)$$

Distances below a safety margin are penalized by

$$\mathcal{L}_{\text{smcl}} = \frac{1}{Z} \sum_{n=1}^N \sum_{t=1}^T m_t w_n \cdot \max(0, \gamma - d_{n,t}), \quad (17)$$

where γ is the safety margin, $m_t \in \{0, 1\}$ is the valid-step mask, and Z normalizes the loss by the effective number of active agents and valid time steps. Here γ is used as a class-agnostic safety proxy that implicitly absorbs the average spatial extent of nearby traffic participants together with an additional clearance buffer, allowing the loss to remain fully differentiable without resorting to box-level geometric collision computation during training. The confidence-aware weight

$$w_n = \max(0, c_n - \epsilon) \quad (18)$$

suppresses low-confidence detections, so the loss focuses on obstacles that are both nearby and reliable. To keep this supervision targeted at planning refinement, the agent trajectories and confidence weights are constructed from stop-gradient motion and detection outputs, i.e.,

$$\tau_{\text{agent}} = f_{\text{motion}}(\text{sg}(\mathbf{M}), \text{sg}(\mathbf{A})), \quad w_n = g(\text{sg}(c_n)), \quad (19)$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator, $\mathbf{M}$ denotes motion predictions, and $\mathbf{A}$ denotes detection anchors. This design keeps the perception and motion branches unchanged while explicitly pushing the refined ego trajectory away from unsafe high-confidence obstacles.

C. Horizon-Aware Regression Weighting

Open-loop planners often match the ground truth well at early steps but drift near the end of the horizon, especially in turning and lane-change scenarios where small early deviations can accumulate into visible terminal errors. Instead of assigning the same weight to every future step, MarginAD reweights only the refined planning regression loss so that terminal trajectory quality receives stronger supervision.

We define a normalized horizon-weight vector $\mathbf{w}_h \in \mathbb{R}^T$ by

$$w_{h,t} = \frac{\tilde{w}_{h,t}}{\frac{1}{T} \sum_{j=1}^T \tilde{w}_{h,j}}, \quad (20)$$

where $\tilde{w}_{h,t}$ is a fixed discrete profile. For the main 3-second setting with $T = 6$, we use a symmetric three-level schedule:

$$\tilde{w}_{h,t} = \begin{cases} a_{\text{edge}}, & t \in \{1, T\}, \\ a_{\text{near}}, & t \in \{2, T-1\}, \\ a_{\text{mid}}, & \text{otherwise}, \end{cases} \quad (21)$$

with $a_{\text{edge}} > a_{\text{near}} > a_{\text{mid}} > 0$, which emphasizes both the near horizon and the terminal horizon while keeping the middle steps relatively lighter. We use the same formulation for the 6-second setting with $T = 12$ in the long-horizon experiments. This low-complexity symmetric design captures the two errors that matter most in our setting, namely immediate safety-critical behavior at early steps and terminal drift at the end of the rollout, without introducing an unrestricted

TABLE I

PLANNING RESULTS ON THE nuSCENES VALIDATION SET UNDER THE VAD OPEN-LOOP PROTOCOL: L2 (M), COLLISION RATE (COL. RATE, %), AND TPC (M) AT DIFFERENT HORIZONS. FOLLOWING [24], EGO-STATUS IS DEACTIVATED FOR ALL METHODS FOR FAIRNESS. “+ MARGIN LOSS” ROWS APPLY THE PROPOSED PLANNING-HEAD REFINEMENT TO THE CORRESPONDING BACKBONE. MISSING ENTRIES FOR DRIVE-WM AND BRIDGEAD REFLECT UNRELEASED WEIGHTS. FPS VALUES ARE MEASURED ON THE ANNOTATED GPUS AND ARE NOT DIRECTLY COMPARABLE ACROSS HARDWARE.

Method	Input	Backbone	L2 (m) ↓				Col. Rate (%) ↓				TPC (m) ↓				FPS ↑
			1s	2s	3s	Avg.	1s	2s	3s	Avg.	1s	2s	3s	Avg.	
Drive-WM [25]	Camera	ResNet101	0.43	0.77	1.20	0.80	0.10	0.21	0.48	0.26	–	–	–	–	–
UniAD [5]	Camera	ResNet101	0.45	0.70	1.04	0.73	0.62	0.58	0.63	0.61	0.41	0.68	0.97	0.68	4.2 (RTX5090)
VAD [6]	Camera	ResNet50	0.41	0.70	1.05	0.72	0.03	0.19	0.43	0.21	0.36	0.66	0.91	0.64	8.6 (RTX5090)
BridgeAD [26]	Camera	ResNet50	0.29	0.57	0.92	0.59	0.01	0.05	0.22	0.09	–	–	–	–	5.0 (RTX3090)
SparseDrive [7]	Camera	ResNet50	0.29	0.58	0.96	0.61	0.01	0.05	0.18	0.08	0.30	0.57	0.85	0.57	10.2 (RTX5090)
SparseDrive + Margin loss	Camera	ResNet50	0.26	0.54	0.92	0.57	0.01	0.05	0.17	0.08	0.27	0.50	0.75	0.51	10.2 (RTX5090)
MomAD [8]	Camera	ResNet50	0.31	0.57	0.91	0.60	0.01	0.05	0.22	0.09	0.30	0.53	0.78	0.54	10.4 (RTX5090)
MarginAD (Ours)	Camera	ResNet50	0.26	0.52	0.87	0.55	0.00	0.05	0.19	0.08	0.26	0.46	0.68	0.47	10.3 (RTX5090)

Reliability. Across five retraining runs on nuScenes, MarginAD obtains Avg. L2 0.55 ± 0.008 , Avg. Col. $0.08 \pm 0.006\%$, and Avg. TPC 0.47 ± 0.012 . The MomAD row is its published single-run value rather than a five-run rerun. The L2 and TPC gains over MomAD are substantially larger than the observed run-to-run deviations, while the collision-rate gain is small because standard nuScenes is already near the collision floor.

TABLE II

PLANNING RESULTS ON THE SHIFT-NUSCENES AND TURNING-NUSCENES VALIDATION SETS UNDER THE VAD [6] PROTOCOL: L2 (M), COLLISION RATE (COL., %), AND TPC (M) AT DIFFERENT HORIZONS. LOWER IS BETTER.

Split	Method	L2 (m) ↓				Col. (%) ↓				TPC (m) ↓			
		1s	2s	3s	Avg.	1s	2s	3s	Avg.	1s	2s	3s	Avg.
Shift-nuScenes	SparseDrive [7]	0.51	0.92	1.39	0.94	0.07	0.24	0.50	0.27	0.76	1.39	2.11	1.42
	MomAD [8]	0.92	1.52	2.16	1.54	0.17	0.34	0.63	0.38	0.73	1.33	2.02	1.36
	MarginAD	0.37	0.72	1.16	0.75	0.00	0.17	0.40	0.19	0.70	1.25	1.88	1.28
Turning-nuScenes	SparseDrive [7]	0.35	0.77	1.46	0.86	0.04	0.17	0.98	0.40	0.34	0.70	1.33	0.79
	MomAD [8]	0.33	0.70	1.24	0.76	0.03	0.13	0.79	0.32	0.32	0.54	1.05	0.63
	MarginAD	0.30	0.65	1.12	0.69	0.08	0.08	0.10	0.09	0.31	0.52	0.83	0.55

Reliability. Across five retraining runs on Shift-nuScenes, SparseDrive obtains Avg. L2/Col./TPC of $0.94 \pm 0.034/0.27 \pm 0.025\%/1.42 \pm 0.038$, MomAD obtains $1.54 \pm 0.031/0.38 \pm 0.023\%/1.36 \pm 0.035$, and MarginAD obtains $0.75 \pm 0.028/0.19 \pm 0.019\%/1.28 \pm 0.031$. The L2 and collision-rate gains over both baselines are statistically significant under a two-sample t -test across the five runs ($p < 0.01$).

learnable weight vector. The refined regression loss is then written as

$$\mathcal{L}_{\text{reg}}^{\text{ref}} = \frac{1}{T} \sum_{t=1}^T w_{h,t} \cdot \|\tau_t - \tau_t^*\|_1, \quad (22)$$

where normalization preserves the overall loss scale.

Training Objective. To make the role of horizon weighting explicit, we write the refined planning loss as

$$\mathcal{L}_{\text{plan}}^{\text{ref}} = \mathcal{L}_{\text{cls}}^{\text{ref}} + \mathcal{L}_{\text{reg}}^{\text{ref}} + \mathcal{L}_{\text{status}}^{\text{ref}}, \quad (23)$$

where horizon-aware weighting is applied only to $\mathcal{L}_{\text{reg}}^{\text{ref}}$. The physical and safety objectives are introduced with fixed balancing coefficients, so that they act as stable regularizers for planning refinement without overwhelming the main imitation objective. The overall training objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{plan}}^{\text{base}} + \mathcal{L}_{\text{plan}}^{\text{ref}} + \mathcal{L}_{\text{phys}} + \lambda_{\text{smcl}} \mathcal{L}_{\text{smcl}}, \quad (24)$$

where $\mathcal{L}_{\text{phys}}$ already includes the internal balancing between steering and curvature violations. In this way, MarginAD improves terminal trajectory quality without changing the decoder structure, while the full objective turns physical feasibility and safety-margin preservation into stable regularizers for safer and more reliable long-horizon planning.

IV. EXPERIMENTS

A. Experimental Setup

Datasets. We conduct extensive open-loop experiments on the widely adopted **nuScenes** [27] dataset, which provides 1,000 driving scenes (700 for training, 150 for validation) with keyframes annotated at 2 Hz and six surround-view RGB images covering a 360° horizontal FOV. Following standard

protocols, we adopt the camera-only end-to-end configuration and use the official splits.

Since standard nuScenes is dominated by simple go-straight routes, we utilize two challenging subsets to better expose planning failures in dynamically constrained scenarios. First, we construct **Shift-nuScenes**, a subset of 579 validation samples across 81 scenes, to focus on traffic-dense interactions with pronounced lateral motion. These samples are mined using deterministic geometry-only criteria: a sample is retained when the future ego displacement satisfies $|\Delta x| > 2.0$ m, $|\Delta y| > 1.5|\Delta x|$, and $|\Delta y| > 3.0$ m, the heading change is below 0.6 rad, and at least two vehicle-category agents are within 20 m. The selection uses only trajectory and agent geometry, independent of any method output or error pattern, and we will release the split tokens and mining script for reproducibility. Second, following MomAD [8], we adopt **Turning-nuScenes** to test temporal stability and terminal drift under sharp heading changes. As introduced in [8], this validation split isolates 680 turning samples across 17 scenes.

Bench2Drive. We use Bench2Drive [28], a benchmark built on CARLA Leaderboard 2.0, for closed-loop evaluation. It covers urban routes with unprotected turns, merges, overtakes, and dense interactions. For fair comparison, we train MarginAD on the official 1,000-clip base set and evaluate on the standard 220 routes.

NAVSIM. We additionally report planning-oriented results on the NAVSIM v1 *navtest* benchmark. Following the official protocol, we report No Collision (NC), Drivable Area Compliance (DAC), Time-To-Collision (TTC), Comfort (Comf.), Ego Progress (EP), and the overall PDMS score. Higher is better

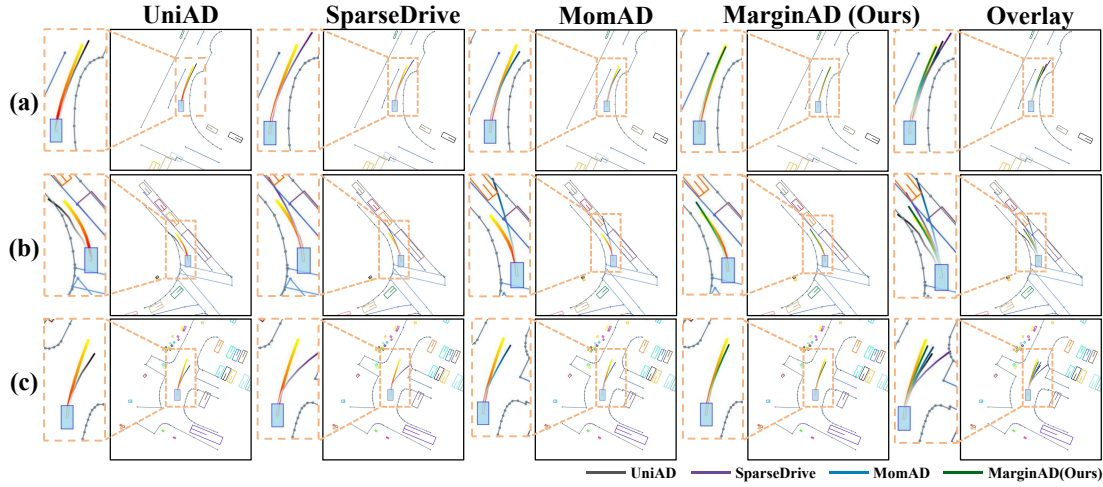


Fig. 3. Qualitative comparison of UniAD, SparseDrive, MomAD, and MarginAD in three representative scenarios: (a) long-horizon drift, (b) unsafe proximity, and (c) unphysical turning. The overlay column shows all predictions together.

for all NAVSIM metrics.

Evaluation Metrics for Planning. We evaluate open-loop planning with three metrics: L2 Displacement Error (L2), Collision Rate (Col.), and Trajectory Prediction Consistency (TPC). Following standard protocols [6], L2 measures deviation from the expert trajectory, Col. evaluates safety, and TPC [8] measures the average discrepancy between predictions from consecutive frames over their overlapping horizons. Following [8], let $\tau_{i,k}^{(t)}$ denote the k -th predicted waypoint for validation sample i obtained at frame t . TPC is then defined as

$$\text{TPC} = \frac{1}{N_T} \sqrt{\sum_{i=1}^{N_T} \sum_{k=1}^T \left\| \tau_{i,k}^{(t)} - \tau_{i,k}^{(t-1)} \right\|_2^2 m_{i,k}}, \quad (25)$$

where N_T is the total number of ground-truth trajectories in the validation set and $m_{i,k} \in \{0, 1\}$ masks steps beyond the overlapping horizon of the two consecutive predictions. As in [8], TPC is a dataset-level statistic. Unless otherwise stated, results are reported for a 3 s horizon with $T = 6$ and $\Delta t = 0.5$ s. For the long-horizon comparison in Table III, we use a 6 s horizon with $T = 12$ and report the 4 s, 5 s, and 6 s metrics. Lower is better for all metrics.

B. Implementation Details

We build MarginAD on MomAD and retrain it in full rather than fine-tuning, keeping the backbone and default optimization settings unchanged unless otherwise specified. Training runs on eight NVIDIA A800 GPUs (80 GB each) and takes about 20 hours. We set $L = 2.8$ m, $\delta_{\max} = 0.9$ rad, $\kappa_{\max} = 0.25 \text{ m}^{-1}$, $\lambda_\delta = \lambda_\kappa = 0.02$, horizon weights [1.3, 1.0, 0.8, 0.8, 1.0, 1.3], safety margin 2.0 m, collision-loss weight 0.05, and confidence threshold 0.3.

C. Main Results

nuScenes. Table I reports the standard open-loop results on nuScenes. MarginAD achieves the best overall performance under the VAD protocol, with an average L2 error of 0.55 m, an average collision rate of 0.08%, and an average TPC of 0.47 m. Compared with MomAD [8], it reduces the averages

from 0.60 m to 0.55 m, from 0.09% to 0.08%, and from 0.54 m to 0.47 m, while remaining consistently better across the three horizons. These results show that explicit kinematic guidance and safety-margin supervision improve trajectory quality, safety, and temporal stability together.

Shift-nuScenes. Table II further evaluates the planner on Shift-nuScenes, our pronounced-lateral-motion subset with denser interactions and weaker local lane guidance. In this harder setting, MarginAD reduces the average L2 error from 1.54 m to 0.75 m, the average collision rate from 0.38% to 0.19%, and the average TPC from 1.36 m to 1.28 m relative to MomAD. MomAD degrades far more than SparseDrive on this split (0.60 m to 1.54 m versus 0.61 m to 0.94 m), yet still attains the lower TPC of the two (1.36 m versus 1.42 m). We attribute both to its momentum prior, which carries the previous plan forward: it resists the abrupt lateral commitment our mining rules select for, leaving predictions laterally late but smooth across frames. The improvement is especially clear at later steps, where the planner must both detour safely and remain stable around nearby vehicles, as also illustrated in Figure 3.

Turning-nuScenes. On Turning-nuScenes, the margin over MomAD is smaller than on Shift-nuScenes but remains consistently across all three metrics. MarginAD improves the average L2 error from 0.76 m to 0.69 m, the average collision rate from 0.32% to 0.09%, and the average TPC from 0.63 m to 0.55 m. MarginAD's 1 s collision rate (0.08%) trails MomAD's (0.03%), but stays nearly flat thereafter (0.08% and 0.10%) while MomAD rises to 0.79%. Since turning scenes place stricter demands on steering smoothness, curvature feasibility, and terminal stability, these gains further support the value of physically feasible planning.

Long Trajectory Prediction. Table III extends the comparison to 4 s, 5 s, and 6 s horizons under the 6 s setting with $T = 12$. On nuScenes, MarginAD improves over MomAD at all long horizons in L2, collision rate, and TPC. On Shift-nuScenes, the long-horizon gains are concentrated in collision rate and TPC, which improve at every horizon, whereas long-range L2 is essentially on par with MomAD (e.g., 1.62 vs. 1.61 at 4 s). This trend is consistent with our horizon-aware weighting design, which aims to suppress terminal drift

TABLE III

COMPARISON OF TRAJECTORY PLANNING PERFORMANCE AT DIFFERENT PREDICTION HORIZONS ON nuSCENES AND SHIFT-nuSCENES. LOWER IS BETTER FOR ALL METRICS.

Split	Method	L2 (m)↓			Col. (%)↓			TPC (m)↓		
		4s	5s	6s	4s	5s	6s	4s	5s	6s
nuScenes	SparseDrive [7]	1.75	2.32	2.95	0.87	1.54	2.33	1.33	1.66	1.99
	MomAD [8]	1.67	1.98	2.45	0.83	1.43	2.13	1.19	1.45	1.61
	MarginAD	1.33	1.84	2.42	0.55	1.02	1.62	1.04	1.31	1.60
Shift-nuScenes	MomAD [8]	1.62	2.17	2.79	0.70	0.72	0.82	2.06	2.53	3.03
	MarginAD	1.61	2.17	2.77	0.65	0.69	0.76	2.01	2.47	2.93

TABLE IV

OPEN-LOOP AND CLOSED-LOOP RESULTS ON BENCH2DRIVE (V0.0.3) UNDER BASE TRAINING SET. “W/O HOR.” REMOVES HORIZON-AWARE WEIGHTING WHILE KEEPING THE PHYSICAL AND SAFETY-MARGIN TERMS UNCHANGED.

Method	Open-loop Metric	Closed-loop Metric			
	Avg. L2↓	DS↑	SR (%)↑	Effi↑	Comf↑
SparseDrive [7]	0.87	44.54	16.71	170.21	48.63
MomAD [8]	0.82	47.91	18.11	174.91	51.20
MarginAD w/o Hor.	0.81	49.24	19.34	173.36	52.72
MarginAD	0.79	49.65	19.52	172.10	53.80

as prediction errors accumulate over time. Figure 4 further illustrates the improved temporal consistency.

Bench2Drive. We further test whether the open-loop gains transfer to closed-loop execution. As shown in Table IV, MarginAD raises the Driving Score from 47.91 to 49.65, the Success Rate from 18.11% to 19.52%, and Comfort from 51.20 to 53.80 over MomAD, with only a minor Efficiency drop (174.91 to 172.10). Removing the horizon-aware weighting (*w/o Hor.*) lowers the Driving Score to 49.24 and the Success Rate to 19.34%, consistent with the horizon term benefiting closed-loop driving and not only open-loop metrics. Dedicated CARLA systems such as Hydra-NeXt [29] (DS 73.86) and TransFuser++ [30] (DS 84.21) score higher because they target closed-loop stack-level driving, whereas MarginAD is a camera-only planner-head refinement and is complementary to such stacks.

NAVSIM. Table V reports performance on NAVSIM v1 *navtest*. Compared with our reproduced MomAD baseline, MarginAD improves NC from 97.7 to 98.1, DAC from 94.6 to 96.3, TTC from 93.1 to 94.6, and PDMS from 84.2 to 86.2, with EP changing from 84.9 to 83.9. Our MomAD reproduction attains the highest EP in the table while trailing the recent entries on NC, DAC, and TTC, a profile that favors progress over clearance; MarginAD trades 1.0 EP point for gains on all three and 2.0 on PDMS, the same trade-off seen in the Efficiency drop on Bench2Drive. These results further show that MarginAD improves planning-oriented safety and compliance metrics under the NAVSIM evaluation protocol.

D. Ablation Study

Table VI validates the role of each component in the proposed planning refinement. The physics-only variant reduces Avg. L2 from 0.60 to 0.57 and Avg. TPC from 0.54 to 0.51, improving trajectory accuracy and temporal stability but without explicitly modeling obstacle clearance. The collision-only variant yields a smaller L2 gain and improves Avg. TPC to 0.52, indicating that safety-margin supervision alone is

TABLE V

PERFORMANCE ON NAVSIM v1 NAVTEST. HIGHER IS BETTER FOR ALL METRICS. * DENOTES OUR REPRODUCTION.

Method	Backbone	NC↑	DAC↑	TTC↑	Comf.↑	EP↑	PDMS↑
VADv2 [31]	ResNet-34	97.2	89.1	91.6	100	76.0	80.9
UniAD [5]	ResNet-34	97.8	91.9	92.9	100	78.8	83.4
Transfuser [3]	ResNet-34	97.7	92.8	92.8	100	79.2	84.0
Hydra-MDP++ [32]	ResNet-34	97.6	96.0	93.1	100	80.4	86.6
FUMP [13]	ResNet-34	98.1	96.2	94.2	100	82.0	87.8
WoTE [33]	ResNet-34	98.5	96.8	94.9	99.9	81.9	88.3
DIVER [11]	ResNet-34	98.5	96.5	94.9	100	82.6	88.3
MomAD* [8]	ResNet-34	97.7	94.6	93.1	100	84.9	84.2
MarginAD	ResNet-34	98.1	96.3	94.6	100	83.9	86.2

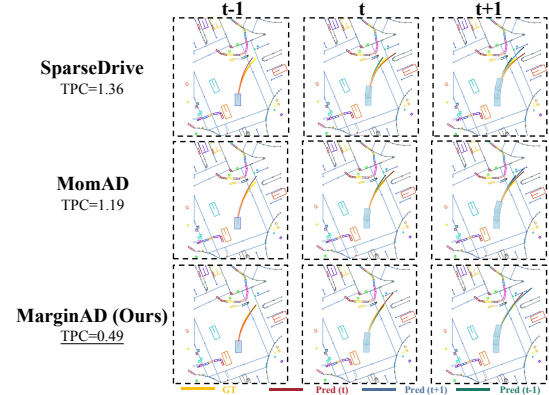


Fig. 4. Qualitative comparison of trajectory prediction consistency (TPC) among SparseDrive, MomAD, and MarginAD across three consecutive frames ($t-1$, t , $t+1$) on nuScenes.

limited while the planner is not yet physically regularized. The horizon-only variant reaches the best Avg. TPC of 0.47, but its Avg. Col. increases to 0.10 because this term suppresses terminal drift rather than spatial proximity. Adding collision supervision on top of physics then reduces Avg. Col. from 0.11 to 0.08. With all three components, MarginAD reaches 0.55/0.08/0.47, the best value on every metric, showing the complementary roles of the three terms: physics regularizes kinematic feasibility, safety-margin supervision enforces clearance, and horizon-aware weighting stabilizes long-horizon prediction.

Hyper-parameter sensitivity. Starting from the collision-only variant, $\gamma = 1.0$ m and $\gamma = 3.0$ m give Avg. L2 of 0.60 and 0.59 against 0.58 at the $\gamma = 2.0$ m used throughout. Starting from the physics-plus-collision variant, replacing the three-level horizon profile by uniform, linear-ramp, exponential-ramp, or learnable weighting gives Avg. TPC of 0.52, 0.50, 0.49, and 0.48 against 0.47 for ours.

E. Analysis of Safety and Physical Feasibility

To verify the physical realism of our planner, we analyze the kinematic statistics of predicted trajectories across the full validation set. As shown in Table VII, although nuScenes expert trajectories (GT) serve as the learning target, they often contain jittery outliers and unphysical spikes due to sensor noise and aggressive human maneuvers, evidenced by the high maximum values in curvature (11.47 m^{-1}) and acceleration (9.74 m/s^2) [27].

Compared to MomAD, MarginAD significantly regularizes the planning output. Specifically, our method reduces the

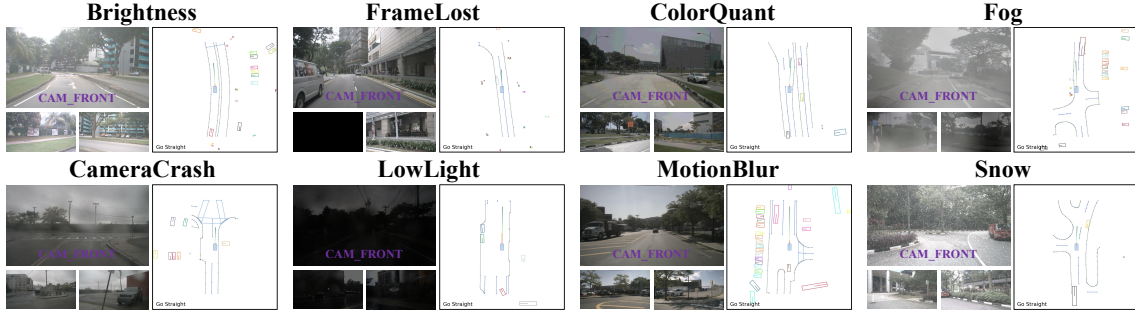


Fig. 5. The eight nuScenes-C corruption types used in our robustness analysis (illumination changes, adverse weather, and sensor/image degradations), each shown with a representative corrupted sample and the corresponding planning trajectory. Within each panel, CAM_FRONT appears above CAM_FRONT_LEFT (bottom left) and CAM_FRONT_RIGHT (bottom right).

TABLE VI

ABLATION STUDY OF THE THREE PROPOSED COMPONENTS ON THE NUSCENES VALIDATION SET. “PHYS.” DENOTES PHYSICS-AWARE KINEMATIC GUIDANCE AND PHYSICAL SUPERVISION, “COLL.” DENOTES THE SAFETY-MARGIN-AWARE COLLISION LOSS, AND “HOR.” DENOTES HORIZON-AWARE WEIGHTING. LOWER IS BETTER FOR ALL METRICS.

Variant	Phys.	Coll.	Hor.	Metric		
				Avg. L2↓	Avg. Col.↓	Avg. TPC↓
Baseline (MomAD)				0.60	0.09	0.54
+ Physics only	Y			0.57	0.11	0.51
+ Collision only		Y		0.58	0.09	0.52
+ Horizon only			Y	0.57	0.10	0.47
+ Physics + Collision	Y	Y		0.57	0.08	0.52
+ Physics + Horizon	Y		Y	0.56	0.09	0.47
Full MarginAD	Y	Y	Y	0.55	0.08	0.47

TABLE VII

COMPARISON OF KINEMATIC STATISTICS BETWEEN GROUND TRUTH (GT), MOMAD [8], AND MARGINAD. OUR METHOD PRODUCES MORE STABLE AND PHYSICALLY FEASIBLE TRAJECTORIES WHILE MAINTAINING EXPERT-LIKE SPEED PROFILES.

Metric	GT (Expert)		MomAD [8]		MarginAD (Ours)	
	Mean ± Std	Max	Mean ± Std	Max	Mean ± Std	Max
Speed (m/s)	5.30 ± 3.51	15.34	5.27 ± 3.45	14.34	5.10 ± 3.32	13.26
Acc. Mag (m/s^2)	0.88 ± 0.97	9.74	0.37 ± 0.36	2.48	0.37 ± 0.35	2.06
Curvature (m^{-1})	0.029 ± 0.30	11.47	0.013 ± 0.07	4.69	0.012 ± 0.05	2.80

maximum curvature from $4.69 m^{-1}$ to $2.80 m^{-1}$ and the maximum acceleration from $2.48 m/s^2$ to $2.06 m/s^2$. Notably, this improvement in smoothness and stability does not come at the cost of performance degradation, as our mean speed (5.10 m/s) remains closely aligned with the expert profile (5.30 m/s). These results demonstrate that our physics-aware kinematic guidance effectively filters out unphysical artifacts, producing trajectories that are more suitable for comfortable downstream execution.

F. Qualitative Results

To evaluate temporal stability, we provide a qualitative comparison of trajectory prediction consistency (TPC) across three consecutive frames ($t-1$, t , $t+1$) in Figure 4. In end-to-end autonomous driving, temporal inconsistency often leads to trajectory jitter, which degrades comfort and can produce unstable control. The TPC values shown in the figure are case-specific measurements for the visualized example rather than dataset-level averages. In this example, SparseDrive and MomAD exhibit larger cross-frame fluctuations, with TPC values of 1.36 and 1.19, respectively, whereas MarginAD reduces the value to 0.49. These visual and quantitative differences

TABLE VIII

ROBUSTNESS COMPARISON UNDER DIFFERENT CORRUPTION TYPES (EASY LEVEL) ON NUSCENES-C. L2 AND TPC IN METERS, COL. IN %; LOWER IS BETTER FOR ALL METRICS.

Method	L2↓	Col.↓	TPC↓	Method	L2↓	Col.↓	TPC↓
Brightness				CameraCrash			
MomAD	0.5729	0.155	0.4936	MomAD	0.6655	0.154	0.5011
MarginAD	0.5546	0.137	0.4780	MarginAD	0.6309	0.154	0.4705
ColorQuant				Fog			
MomAD	0.5731	0.118	0.4936	MomAD	0.5764	0.147	0.4981
MarginAD	0.5566	0.104	0.4733	MarginAD	0.5590	0.139	0.4745
FrameLost				LowLight			
MomAD	0.7086	0.348	0.6577	MomAD	0.5982	0.192	0.4974
MarginAD	0.6572	0.334	0.5978	MarginAD	0.5770	0.144	0.4808
MotionBlur				Snow			
MomAD	0.5764	0.150	0.5066	MomAD	0.7233	0.178	0.5483
MarginAD	0.5643	0.128	0.4897	MarginAD	0.7471	0.130	0.5315

indicate that MarginAD produces more coherent predictions across frames and better suppresses erratic trajectory shifts.

G. Robustness Analysis

To evaluate robustness under perceptual degradation, we follow the nuScenes-C corruption benchmark [34], [35]. As shown in Table VIII, MarginAD improves all three metrics over MomAD on six of the eight corruptions, and improves L2 and TPC while matching the collision rate on CameraCrash. Averaged over all eight corruptions, it reduces L2 from 0.6243 to 0.6058, the collision rate from 0.1802% to 0.1588%, and TPC from 0.5245 to 0.4995. The advantage is most consistent on corruptions that weaken scene continuity or visual fidelity (e.g., FrameLost, LowLight, MotionBlur), showing that the gain is not limited to clean inputs. Snow is the only case with a slight L2 increase, yet MarginAD still attains a lower collision rate and better temporal consistency.

Figure 5 shows representative corrupted cases with the resulting trajectories. Consistent with Table VIII, MarginAD exhibits less long-horizon drift and safer clearance when scene cues become unreliable, matching its design: physics-aware guidance regularizes the trajectory and the safety-margin loss keeps discouraging unsafe plans under corrupted inputs.

V. CONCLUSION

We presented MarginAD, an end-to-end planning framework that refines the planning head with physics-aware kinematic guidance, a safety-margin-aware collision loss,

and horizon-aware regression weighting. Across open-loop nuScenes and its Shift-nuScenes subset, closed-loop Bench2Drive, and NAVSIM, MarginAD consistently improves trajectory accuracy, safety, and temporal stability over the MomAD baseline, showing that injecting explicit physical and safety structure into planning refinement is effective for trajectory-based driving systems. Future work will explore uncertainty-aware safety modeling, adaptive safety margins, and tighter coupling between prediction confidence and planning refinement.

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